

Section 15.8: Triple Integrals in Spherical Coordinates

Multiple Integrals

MTH 310
Calculus III

Why Spherical Coordinates?

Cylindrical coordinates handle solids with circular cross-sections in the xy -plane. But some solids are more naturally described by their **distance from the origin**:

- Spheres: $x^2 + y^2 + z^2 = a^2$
- Cones: $z = c\sqrt{x^2 + y^2}$
- Hemispheres, spherical shells, ice-cream-cone regions

In cylindrical coordinates, a sphere is $r^2 + z^2 = a^2$ — still messy. In **spherical coordinates**, it becomes simply $\rho = a$.

Spherical coordinates describe a point by:

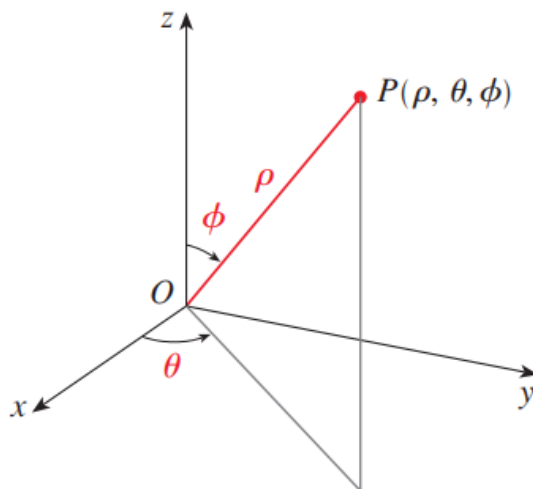
1. How far it is from the **origin** (ρ)
2. What direction it's in (θ and ϕ)

Use spherical when the solid or integrand involves $x^2 + y^2 + z^2$ (distance from origin). Use cylindrical when it involves $x^2 + y^2$ (distance from the z -axis).

The Spherical Coordinate System

A point in 3D space is described by (ρ, θ, ϕ) where:

- ρ = distance from the **origin** ($\rho \geq 0$)
- θ = angle in the xy -plane from the positive x -axis (same as in cylindrical/polar)
- ϕ = angle from the positive z -axis down to the point ($0 \leq \phi \leq \pi$)



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Important: ϕ is measured from the **top** (positive z -axis):

- $\phi = 0$: pointing straight up (north pole)
- $\phi = \pi/2$: in the xy -plane (equator)
- $\phi = \pi$: pointing straight down (south pole)

ρ tells you how far, θ tells you which compass direction, ϕ tells you how far from vertical. Think of it as GPS coordinates on a sphere: longitude (θ) and colatitude (ϕ).

Coordinate Surfaces: What Does Each Constant Look Like?

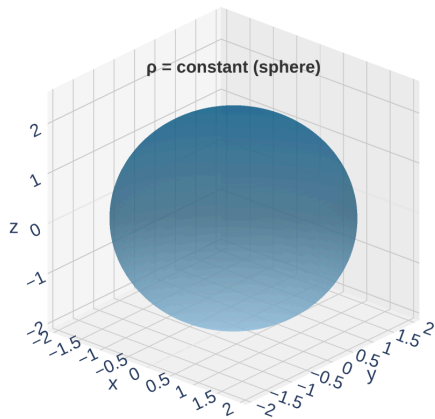
What surface do you get when you hold one coordinate constant?

Equation	Surface	Description
$\rho = c$	Sphere	Centered at origin, radius c

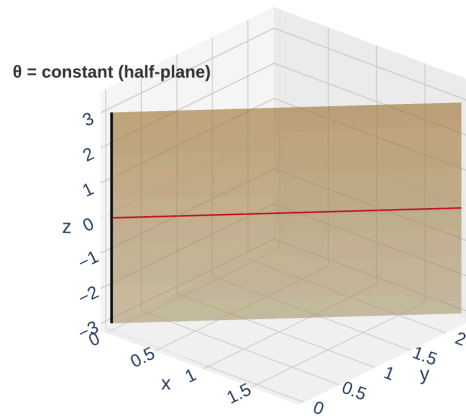
$\theta = c$	Half-plane	Vertical, hinged on the z -axis
$\phi = c$	Cone	Half-cone opening from the z -axis

Some important special cases for ϕ :

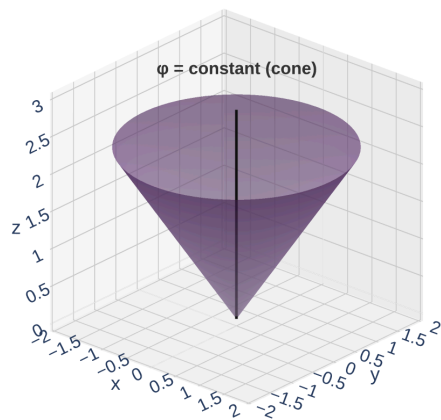
- $\phi = 0$: the positive z -axis (a degenerate “cone” — just a ray)
- $\phi = \pi/4$: a cone at 45 degrees from vertical
- $\phi = \pi/2$: the xy -plane (the “widest” cone — a flat disk)
- $\phi = \pi$: the negative z -axis



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$\rho = \text{const}$ gives a sphere, $\theta = \text{const}$ gives a half-plane, $\phi = \text{const}$ gives a cone. These are the building blocks for spherical regions.

Conversion Formulas

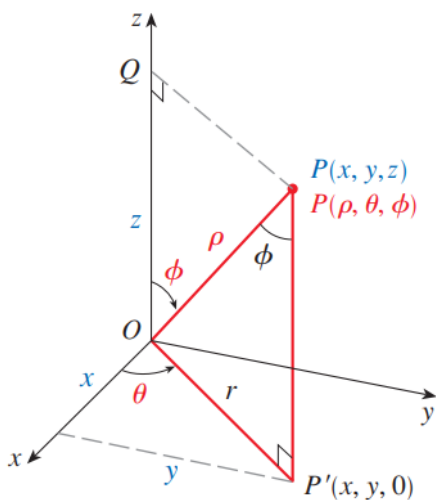
Spherical $\leftrightarrow$ Cartesian Conversions

Spherical to Cartesian:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

Cartesian to Spherical:

$$\rho^2 = x^2 + y^2 + z^2, \quad \tan \theta = \frac{y}{x}, \quad \cos \phi = \frac{z}{\rho}$$



Useful relationship to cylindrical:

$$r = \rho \sin \phi, \quad z = \rho \cos \phi$$

This makes sense: the projection of ρ onto the xy -plane is $\rho \sin \phi$ (the cylindrical r), and the vertical component is $\rho \cos \phi$ (the height z).

Common conversions:

Cartesian	Spherical	Surface
$x^2 + y^2 + z^2 = 16$	$\rho = 4$	Sphere of radius 4
$z = \sqrt{x^2 + y^2}$	$\phi = \pi/4$	Cone at 45 degrees
$z = 0$	$\phi = \pi/2$	The xy -plane
$x^2 + y^2 + z^2 = 2z$	$\rho = 2 \cos \phi$	Sphere tangent to origin

The master identity is $\rho^2 = x^2 + y^2 + z^2$. For cones $z = c\sqrt{x^2 + y^2}$, divide both sides by ρ to get $\cos \phi = c \sin \phi$, i.e., $\phi = \arctan(1/c)$.

Quick Check: Convert Equations to Spherical

Write each equation in spherical coordinates:

1. $x^2 + y^2 + z^2 = 16$

2. $z = \sqrt{x^2 + y^2}$

For (1), use $\rho^2 = x^2 + y^2 + z^2$. For (2), write in terms of ρ , ϕ and simplify.

The Volume Element in Spherical Coordinates

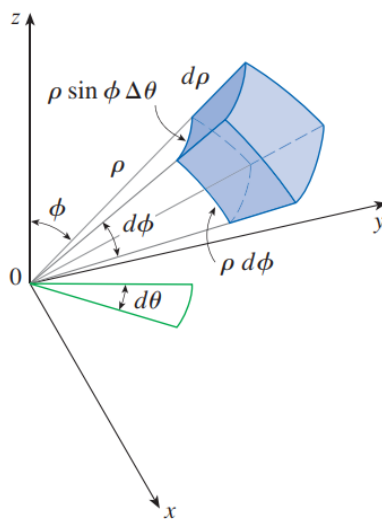
In Cartesian, $dV = dx dy dz$. In cylindrical, $dV = r dz dr d\theta$. What about spherical?

A small “spherical wedge” at position (ρ, θ, ϕ) has three edge lengths:

- Radial: $d\rho$
- Longitudinal arc: $\rho \sin \phi d\theta$ (circle of radius $\rho \sin \phi$ in the xy -plane)
- Latitudinal arc: $\rho d\phi$ (arc along a meridian)

The volume of this wedge is the product of the three edge lengths:

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta$$



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The magnification factor $\rho^2 \sin \phi$ accounts for the stretching of the coordinate grid. It's the price of using curved coordinates — but it's worth it when the limits become simple constants.

Quick Check: Describe the Region

What solid does the following region describe?

$$0 \leq \rho \leq 3, \quad 0 \leq \phi \leq \pi/2, \quad 0 \leq \theta \leq 2\pi$$

Think about each limit: what does $\rho \leq 3$ mean? What does $\phi \leq \pi/2$ restrict? What does θ from 0 to 2π mean?

When to Use Spherical Coordinates

Use spherical when:

1. The region is bounded by **spheres** ($\rho = a$) and/or **cones** ($\phi = c$)
2. The integrand involves $x^2 + y^2 + z^2$ (becomes ρ^2)
3. The solid is a full ball, hemisphere, spherical shell, or “ice-cream cone” shape

Template:

$$\iiint_E f dV = \int_{\alpha}^{\beta} \int_{\phi_1}^{\phi_2} \int_{\rho_1(\theta, \phi)}^{\rho_2(\theta, \phi)} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\phi d\theta$$

Typical order: $d\rho d\phi d\theta$. The ρ -integral is innermost (simplest limits), and θ is outermost (usually 0 to 2π).

The decision tree: **circles in xy -plane?** $\rightarrow$ cylindrical. **Spheres or cones from origin?** $\rightarrow$ spherical. **Neither?** $\rightarrow$ Cartesian.

Example 1: Integral Over a Lower Hemisphere

Example 1

Evaluate $\iiint_H (9 - x^2 - y^2) dV$, where H is the solid lower hemisphere $x^2 + y^2 + z^2 \leq 9, z \leq 0$.

Convert to spherical. What are the limits on ρ, ϕ, θ ? Remember $x^2 + y^2 = \rho^2 \sin^2 \phi$ and $dV = \rho^2 \sin \phi d\rho d\phi d\theta$.

Example 2: Between Spheres, Above a Cone

Example 2

Evaluate $\iiint_E xyz \, dV$, where E lies between the spheres $\rho = 2$ and $\rho = 4$, and above the cone $\phi = \pi/3$.

Convert xyz to spherical coordinates and set up the integral. What are the limits on ρ , ϕ , θ ?

Choosing the Right Coordinate System

After Chapters 15.6–15.8, you have three coordinate systems for triple integrals. How do you choose?

Coordinate System	Best When	Volume Element
Cartesian (x, y, z)	Rectangular solids, no obvious symmetry	$dV = dx \, dy \, dz$
Cylindrical (r, θ, z)	Circular symmetry about the z -axis: cylinders, paraboloids, cones	$dV = r \, dz \, dr \, d\theta$
Spherical (ρ, θ, ϕ)	Symmetry about the origin: spheres, cones from origin	$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

Quick decision guide:

- See $x^2 + y^2$ in the region or integrand? → **Cylindrical**
- See $x^2 + y^2 + z^2$ in the region or integrand? → **Spherical**
- Region described by planes only? → **Cartesian**
- Sphere + cone? → **Spherical** (both become constants)
- Cylinder + horizontal planes? → **Cylindrical**

The “right” coordinate system is the one that makes the **limits of integration** simple. If the boundaries become constants, you've found the right system.

Homework

Section 15.8: 1, 3, 5, 6, 7, 9, 11, 15, 17, 19, 20, 23, 25, 27, 29, 31, 37, 43, 50